

BALASORE, India

WMO: 428950

Lat: 21.517N

Lon: 86.933E

Elev: 66

StdP: 14.66

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.7	57.7	42.4	40.1	72.0	45.2	44.6	68.3	9.3	70.7	7.7	70.5	2.0	0	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.5	99.7	80.6	97.2	80.6	95.2	80.7	85.0	93.8	84.1	92.9	83.3	91.7	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.7	171.0	91.2	81.8	165.9	90.4	81.0	161.3	89.2	49.7	94.3	48.6	93.1	47.6	92.0	93.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	7.7	7.1	DB	51.6	105.6	1.8	4.3	50.4	108.7	49.3	111.3	48.3	113.7	47.0	116.9	(4)
(5)			WB	47.5	87.3	1.5	2.3	46.4	89.0	45.5	90.4	44.7	91.7	43.6	93.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	69.4	75.3	82.3	86.4	87.5	86.5	84.7	84.4	84.1	82.0	76.4	70.4	(6)	
	DBStd	6.81	3.20	3.93	3.36	3.79	3.34	3.44	2.23	2.00	2.33	2.73	3.11	3.06	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	6	4	0	0	0	0	0	0	0	0	0	0	2	(9)	
	CDD50	11246	603	708	1001	1093	1162	1096	1077	1066	1023	993	792	632	(10)	
	CDD65	5775	141	288	536	643	697	646	612	601	573	528	342	168	(11)	
	CDH74	66077	733	2064	5725	8356	9653	8694	7597	7341	6868	5629	2526	891	(12)	
	CDH80	29415	106	661	2507	4428	5408	4477	3214	2959	2716	2071	744	124	(13)	
(14) Wind	WSAvg	2.5	2.0	2.2	2.9	3.8	3.6	3.0	2.6	2.4	2.0	1.7	1.8	1.9	(14)	
(15) Precipitation	PrecAvg	64.3	0.9	1.2	1.1	2.4	5.2	10.6	11.9	11.8	11.2	6.7	1.2	0.2	(15)	
	PrecMax	84.8	4.2	6.0	7.3	5.4	15.8	23.9	22.2	21.4	22.3	26.3	8.9	1.7	(16)	
	PrecMin	44.4	0.1	0.0	0.0	0.0	0.5	3.9	4.9	4.7	1.7	0.0	0.0	0.0	(17)	
	PrecStd	11.2	1.1	1.7	1.5	1.8	3.1	4.1	3.9	3.7	5.1	6.3	1.8	0.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.1	93.6	98.4	103.7	103.3	102.3	95.3	93.6	93.8	93.4	89.8	85.7	(19)	
		MCWB	66.0	70.4	74.5	78.6	80.4	83.2	83.0	82.2	81.5	79.2	73.0	68.2	(20)	
	2%	DB	82.6	89.8	95.1	99.7	99.2	97.8	92.8	91.8	92.2	91.6	87.8	83.1	(21)	
		MCWB	65.6	70.0	74.1	79.0	82.4	83.4	82.2	81.8	81.2	79.0	73.0	66.9	(22)	
	5%	DB	80.5	86.9	92.8	96.9	96.6	95.2	90.9	90.4	90.9	90.1	86.2	81.2	(23)	
		MCWB	64.7	69.3	74.0	79.3	82.2	83.1	81.5	81.4	80.9	78.4	72.2	65.7	(24)	
	10%	DB	78.3	84.4	90.7	94.5	94.5	93.1	89.4	89.0	89.4	88.5	84.5	79.2	(25)	
		MCWB	63.8	69.0	73.9	79.2	82.2	82.4	81.1	81.0	80.7	77.7	71.0	64.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.8	76.8	80.9	84.1	86.2	86.2	84.2	83.7	83.6	81.9	78.1	72.4	(27)	
		MCDB	79.3	85.4	91.2	95.0	95.1	94.9	91.8	90.5	90.7	90.0	85.5	80.3	(28)	
	2%	WB	69.7	74.8	79.4	82.8	85.2	85.0	83.0	82.7	82.5	80.9	76.3	70.1	(29)	
		MCDB	77.1	83.0	88.9	93.7	94.1	93.7	90.2	89.6	89.7	88.5	83.4	78.3	(30)	
	5%	WB	68.3	73.4	78.4	81.9	84.3	84.0	82.3	81.9	81.7	80.1	75.2	68.6	(31)	
		MCDB	76.1	81.0	87.1	92.4	93.3	92.7	89.2	88.5	88.6	87.2	82.5	77.1	(32)	
	10%	WB	66.7	72.1	77.3	81.0	83.4	83.0	81.6	81.4	81.0	79.4	73.8	67.0	(33)	
		MCDB	74.5	79.5	85.6	91.1	92.2	91.4	88.0	87.6	87.5	85.9	80.6	75.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	18.9	17.9	17.4	15.5	11.7	9.0	8.8	9.5	12.4	17.1	19.6	(35)	
		MCDBR	21.9	21.4	19.7	19.7	17.7	15.5	11.7	10.9	11.8	14.1	17.6	20.6	(36)	
	5% WB	MCWBR	8.8	8.0	7.6	7.6	8.1	7.1	5.0	4.7	4.8	5.6	7.1	8.6	(37)	
		MCWBR	17.8	16.8	17.0	17.1	15.7	14.6	10.4	9.9	10.8	11.8	13.8	17.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.575	0.557	0.610	0.624	0.666	0.686	0.594	0.572	0.567	0.580	0.612	0.619	(40)	
		taud	1.764	1.843	1.773	1.795	1.745	1.713	1.925	1.999	1.998	1.905	1.753	1.700	(41)	
	Ebn at Noon		218	235	229	228	218	211	232	238	237	226	207	200	(42)	
		Edh at Noon	66	64	71	71	74	76	61	57	56	59	66	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		1300	1571	1820	1998	1950	1489	1280	1338	1348	1418	1337	1207	(44)	
	RadStd		72	95	85	81	99	115	110	89	79	130	78	99	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.59	+0.68	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	+0.72	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page